Effective Utility Services, LLC

Reducing the Cost of Energy to Schools

BRECKSVILLE-BROADVIEW HTS. CITY SCHOOL DISTRICT

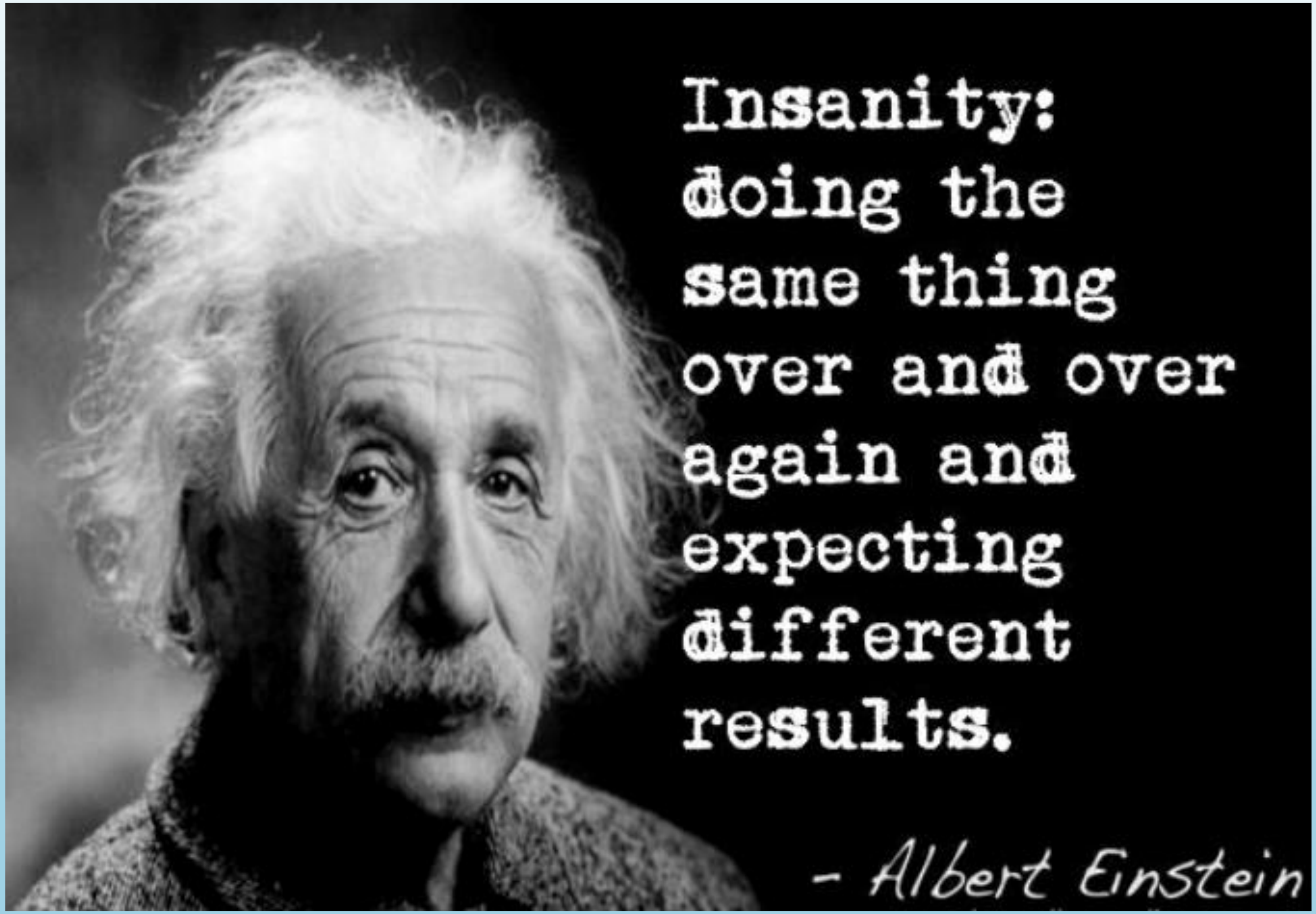
"where fine education is a heritage"

Michael D. Spacek, PE, CEM LEED AP
Director of Energy Services
Engineered Process Systems, Ltd.
PO Box 471 – 205 Sprowl Road
Huron, OH 44839
PH 419-433-7048
Mobile 419-602-0503
mike@bestuseofenergy.com

EPS ENGINEERED PROCESS SYSTEMS, LTD.

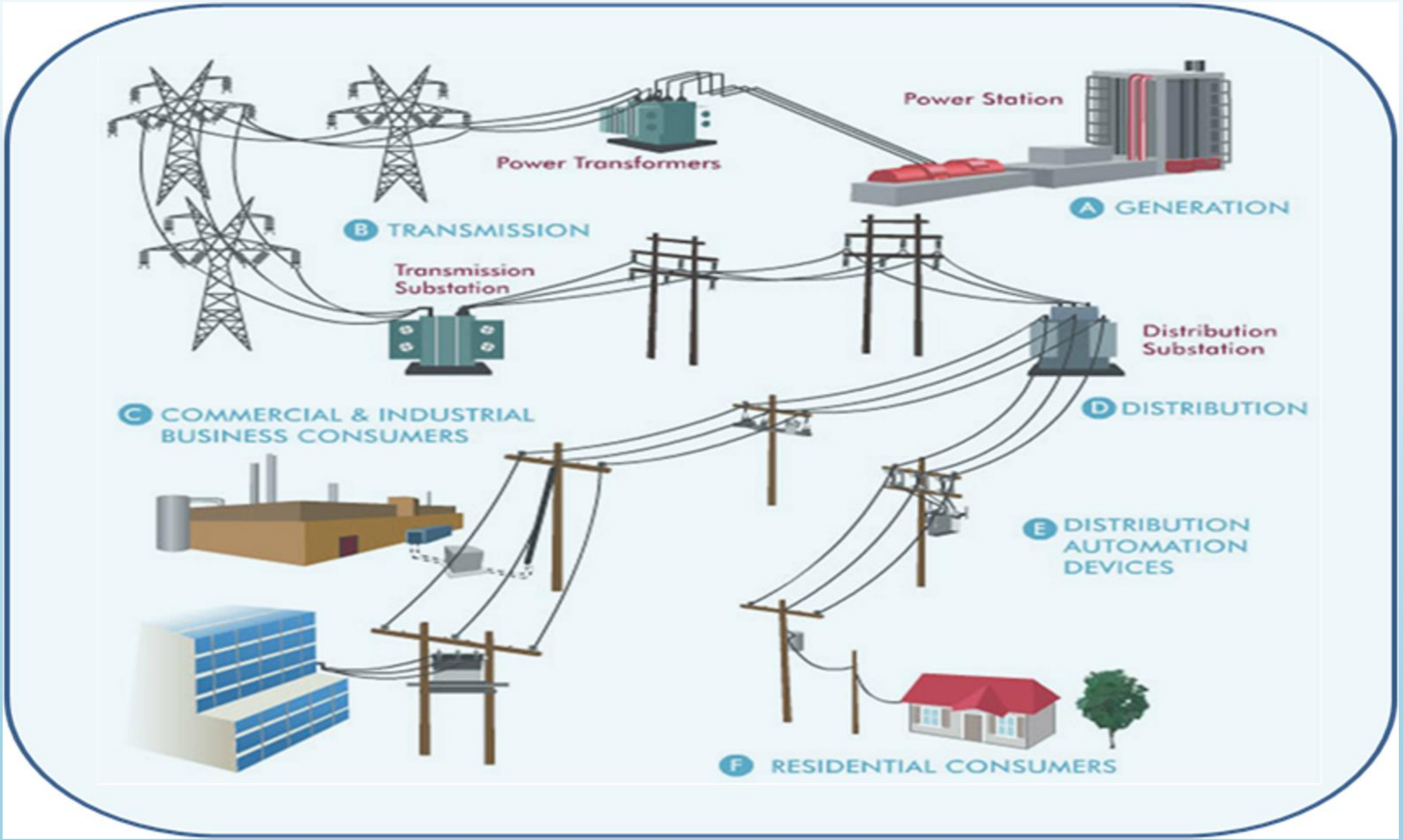
Energy

- $E=MC^2$
- For our discussion, we will focus on the cost of electrical energy from The Illuminating Company
- We will show how to calculate an electric bill
- We will show various ways to lower electric costs

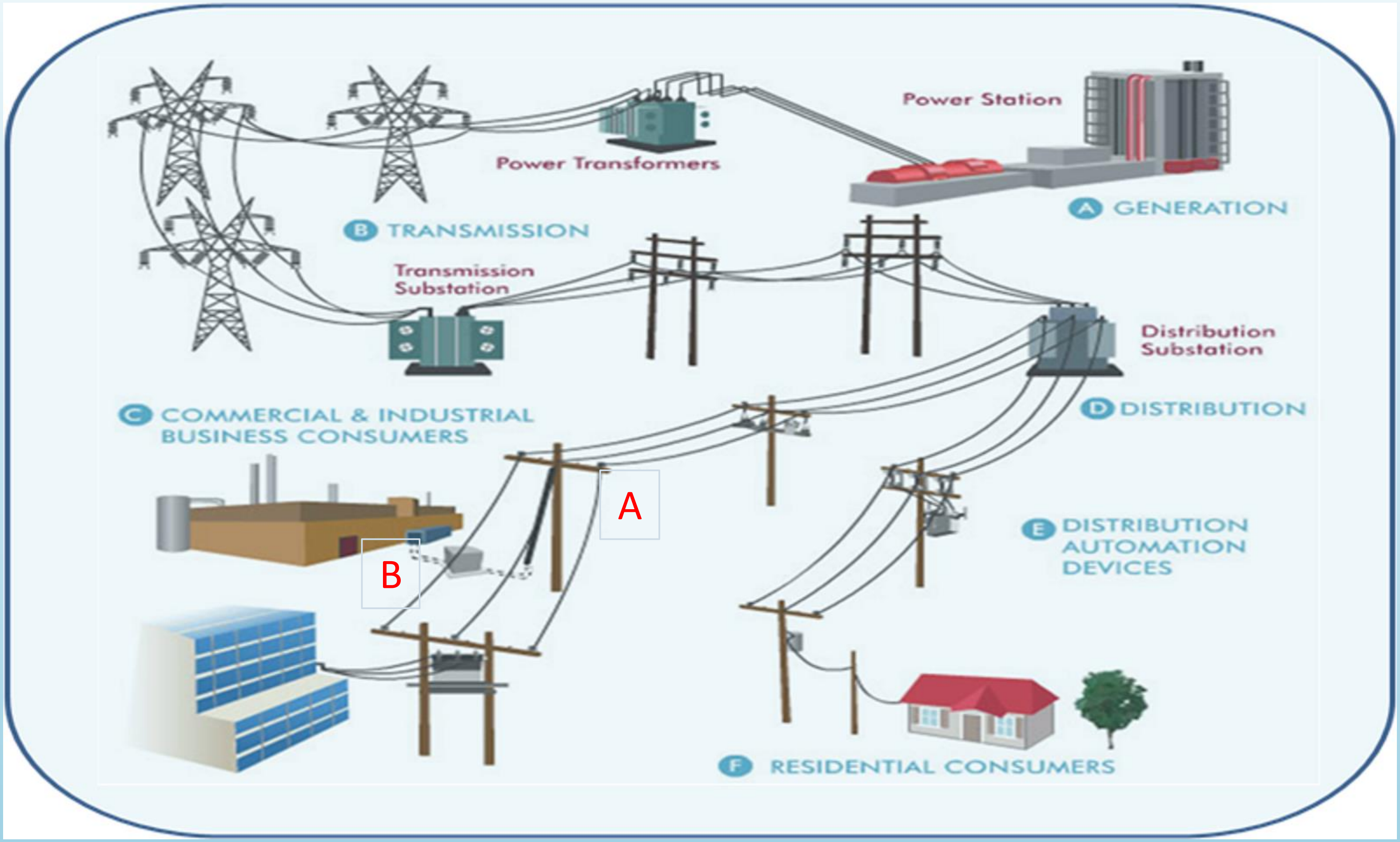

Insanity:
doing the
same thing
over and over
again and
expecting
different
results.
- Albert Einstein

EUS Has Reduced BBHCS Electrical Costs Approximately $270,000 and Growing

- Started EUS Energy Cost Reduction Program 8/1/14
- Utilize CEI Filed Rates
  - Available rates include GS, GP, GSUB and GT
- Reconfigured Campus electrical distribution system to qualify for lower rate
- Prior to 8/14 campus had six GS meters
- EUS combined the campus to one GP meter
- Saves approximately $5,400/month
- Approximately 10/22 savings increase approximately 50%
- On 8/24 savings increase to approximately $120,000 per year



The Electric Delivery System



EUS Changes the Delivery Point

THE ILLUMINATING COMPANY ENERGY ANALYSIS REPORT

Account Number: 110 062 761 240 **Service Address:** 8611 WIESE RD BRECKSVILLE OH 44141
Premise: 1630000646 **Portion:** C10 **Service From:** 11/29/2018 **Service To:** 12/28/2018 **Billing Days:** 30 **Avg KWH/Day:** 601
Rate: General Service Secondary **Measured Load:** 64.1 **KVAR:** 17.5

Meter Number	Previous MR	Current MR	Difference	Type of Reading	Multiplier
5330175353	6,036	6,487	18,040 KWH	Actual	40

Customer Number: 0803750840 1630000646 - General Service Secondary - CE-GSD

Customer Charge	7.00
Distribution Related Component	876.45
Cost Recovery Charges	352.79
School Distribution Credit	-40.87
Generation Related Component	934.47
Present Electric:	2,129.84

Account Number: 110 062 763 872 **Service Address:** 6380 MILL RD HI SCH BRECKSVILLE OH 44141
Premise: 1470095564 **Portion:** C10 **Service From:** 12/05/2018 **Service To:** 01/03/2019 **Billing Days:** 30 **Avg KWH/Day:** 10,570
Rate: General Service Primary **Metered Load:** 999.3 **KVAR:** 358.1

Meter Number	Previous MR	Current MR	Difference	Type of Reading	Multiplier
173606	15,139	15,139	0 KWH	Actual	1
5314121622	22,651	23,708	317,100 KWH	Actual	300

Customer Number: 0803750840 1470095564 - General Service Primary - CE-GPD

Customer Charge	150.00
Distribution Related Component	6,023.38
Cost Recovery Charges	8,338.87
School Distribution Credit	-235.75
Generation Related Component	16,425.78
Present Electric:	30,702.28

Account Number: 110 063 011 552 **Service Address:** 9457 HIGHLAND DR BRECKSVILLE OH 44141
Premise: 1520000639 **Portion:** C10 **Service From:** 11/20/2018 **Service To:** 12/19/2018 **Billing Days:** 30 **Avg KWH/Day:** 641
Rate: General Service Secondary **Metered Load:** 75.5 **KVAR:** 9.6

Meter Number	Previous MR	Current MR	Difference	Type of Reading	Multiplier
9062500017	37,138	37,500	14,480 KWH	Actual	40
2806684	652,050	656,785	4,735 KWH	Actual	1

Customer Number: 0803750840 1520000639 - General Service Secondary - CE-GSD

Customer Charge	7.00
Distribution Related Component	1,012.25
Cost Recovery Charges	407.94
School Distribution Credit	-48.04
Generation Related Component	995.34
Present Electric:	2,374.49

Account Number: 110 063 061 219 **Service Address:** 27 PUBLIC SQ BRECKSVILLE OH 44141
Premise: 1000000649 **Portion:** C10 **Service From:** 11/28/2018 **Service To:** 12/27/2018 **Billing Days:** 30 **Avg KWH/Day:** 725
Rate: General Service Secondary **Metered Load:** 133.4 **KVAR:** 78.8

Meter Number	Previous MR	Current MR	Difference	Type of Reading	Multiplier
5311188051	6,704	6,750	7,360 KWH	Estimate	160
9103776612	28,161	28,341	14,400 KWH	Estimate	80

Customer Number: 0803750840 1000000649 - General Service Secondary - CE-GSD

Customer Charge	7.00
Distribution Related Component	1,710.14
Cost Recovery Charges	675.35
School Distribution Credit	-87.85
Generation Related Component	1,127.17
Present Electric:	3,431.81

- Without the EUS Program
- Billed Load 1,270-kW
- Amount $24,500

Master Invoice: 90006848893 Master Account: 210 000 516 190 Page 1 of 1

THE ILLUMINATING COMPANY SUMMARY ACCOUNT DETAIL LIST - Master Account: 210 000 516 190

Summary Account: 210 000 516 190	**Current Month Bill Amount:** $ 38,638.42	**Total Accounts:** 6	
Customer Name: WILKES & CO	**Total Amount Due:** $ 38,638.42	**Total Billed:** 4	
Billing Date: January 15, 2019	**Due Date:** February 19, 2019	**Total KWH:** 376,115	

Bill Type Codes: A=Actual C=Customer E=Estimate N=No Bill

Account Number	Service Address	Rate Category	Billing Period	Amount	Price to Compare	Billed KWH	Billed Load	Billed KVAR	Bill Type
110 062 761 249	8611 WIESE RD BRECKSVILLE OH 44141								
		CE-GSD	11/29/2018 12/28/2018	$1,195.37		18,040	64.1	17.5	A
		FES-8118	11/29/2018 12/28/2018	$934.47		18,040	64.1	35.0	A
110 062 761 322	6376 MILL RD BROADVIEW HEIGHTS OH 44147								N
110 062 763 799	6812 MILL RD BRECKSVILLE OH 44141								N
110 062 763 872	6380 MILL RD HI SCH BRECKSVILLE OH 44141								
		CE-GPD	12/05/2018 01/03/2019	$14,276.50		317,100	999.3	398.1	A
		FES-8118	12/05/2018 01/03/2019	$16,425.78		317,100	999.3	796.2	A
110 063 011 552	9457 HIGHLAND DR BRECKSVILLE OH 44141								
		CE-GSD	11/20/2018 12/19/2018	$1,379.15		19,215	75.5	9.6	A
		FES-8118	11/20/2018 12/19/2018	$995.34		19,215	75.5	19.2	A
110 063 061 219	27 PUBLIC SQ BRECKSVILLE OH 44141								
		CE-GSD	11/28/2018 12/27/2018	$2,304.64		21,760	133.4	78.8	E
		FES-8118	11/28/2018 12/27/2018	$1,127.17		21,760	133.4	157.6	E

How Did We Do It?

- Bills are a combination of Demand (kW) and Energy (kWh) and Fixed Charges ($$)
- Comprised of Rate Schedule and Riders

GENERAL SERVICE - SECONDARY (RATE "GS")

AVAILABILITY:

Available to general service installations requiring Secondary Service. Secondary Service is defined in the Company's Electric Service Regulations. Choice of voltage shall be at the option of the Company.

SERVICE:

All service under this rate schedule will be served through one meter for each installation.

RATE:

All charges under this rate schedule shall be calculated as described below and charged on a monthly basis.

Distribution Charges:	
Service Charge:	$7.00
Capacity Charge:	
Up to 5 kW of billing demand	$13.6800
For each kW over 5 kW of billing demand	$7.4790
Reactive Demand Charge applicable to three phase customers only	
For each rkVA of reactive billing demand	$0.36

BILLING DEMAND:

The billing demand for the month shall be the greatest of:

1. Measured Demand, being the highest thirty (30) minute integrated kW
2. 5.0 kW
3. The Contract Demand

Measured Demand shall be estimated for all customers not having a demand meter and using over 1,000 kWh per month by applying a factor of 200 by the following formula: Measured Demand = kWh / 200.

REACTIVE BILLING DEMAND:

For installations metered with reactive energy metering, the reactive billing demand in rkVA for the month shall be determined by multiplying the Measured Demand by the ratio of the measured lagging reactive kilovoltampere hours to the measured kilowatthours by the following formula: rkVA = Measured Demand X (measured lagging reactive kilovoltampere hours ÷ measured kilowatthours). For all other installations, the reactive billing demand shall be the integrated reactive demand occurring coincident with the Measured Demand.

GENERAL SERVICE - PRIMARY (RATE "GP")

AVAILABILITY:

Available to general service installations requiring Primary Service. Primary Service is defined in the Company's Electric Service Regulations. Choice of voltage shall be at the option of the Company.

SERVICE:

All service under this rate schedule will be served through one meter for each installation.

The customer will be responsible for all transforming, controlling, regulating and protective equipment and its operation and maintenance.

RATE:

All charges under this rate schedule shall be applied as described below and charged on a monthly basis.

Distribution Charges:	
Service Charge:	$150.00
Capacity Charge:	
For each kW of billing demand	$2.4050
Reactive Demand Charge applicable to three phase customers only	
For each rkVA of reactive billing demand	$0.36

BILLING DEMAND:

The billing demand for the month shall be the greatest of:

1. Measured Demand, being the highest thirty (30) minute integrated kW
2. 30.0 kW
3. The Contract Demand

REACTIVE BILLING DEMAND:

For installations metered with reactive energy metering, the reactive billing demand in rkVA for the month shall be determined by multiplying the Measured Demand by the ratio of the measured lagging reactive kilovoltampere hours to the measured kilowatthours by the following formula: rkVA = Measured Demand X (measured lagging reactive kilovoltampere hours ÷ measured kilowatthours). For all other installations, the reactive billing demand shall be the integrated reactive demand occurring coincident with the Measured Demand.

SUMMARY RIDER

Rates and charges included in the rate schedules listed in the following matrix shall be modified consistent with the terms and conditions of the indicated Riders:

		Rate Schedule							
	Rider - (Sheet)	RS	GS	GP	GSU	GT	STL	TRF	POL
	Residential Distribution Credit - (81)	●							
A	Transmission and Ancillary Services - (83)	●	●	●	●	●	●	●	●
	School Distribution Credit - (85)		●	●	●				
	Business Distribution Credit - (86)		●	●					
P	Universal Service - (90)	●	●	●	●	●	●	●	●
P	Energy Efficiency - (91)	●	●	●	●	●			
	State kWh Tax - (92)	●	●	●	●	●	●	●	●
	Net Energy Metering - (93)	●	●	●	●	●			
T	Demand Side Management - (97)	●							
Q	Distribution Uncollectible - (99)	●	●	●	●	●	●	●	●
	Deferred Transmission Cost Recovery - (100)	●	●	●	●	●	●	●	●
	Green Resource - (104)	●	●	●	●	●	●	●	●
	Advanced Metering Infrastructure / Modern Grid - (106)	●	●	●	●	●	●	●	●
	Delivery Service Improvement - (108)	●	●	●	●				
Q	PIPP Uncollectible Recovery - (109)	●	●	●	●	●	●	●	●
T	Demand Side Management and Energy Efficiency - (115)	●	●	●	●	●	●	●	●

● - Rider is applicable or available to the Rate Schedules indicated

A - Rider is updated/reconciled annually
Q - Rider is updated/reconciled quarterly
T - Rider is updated/reconciled twice per year
P - Rider is updated/reconciled periodically

Years ago the Riders were insignificant

SUMMARY RIDER

Rates and charges included in the rate schedules listed in the following matrix shall be modified consistent with the terms and conditions of the indicated Riders:

		Rate Schedule							
	Rider - (Sheet)	RS	GS	GP	GSU	GT	STL	TRF	POL
Q	Advanced Metering Infrastructure / Modern Grid - (106)	●	●	●	●		●	●	●
Q	Alternative Energy Resource - (84)	●	●	●	●	●	●	●	●
	Business Distribution Credit - (86)		●	●					
Q	CEI Delta Revenue Recovery - (112)	●	●	●	●	●	●	●	●
	Commercial High Load Factor Experimental TOU – (130)		●	●					
	Deferred Fuel Cost Recovery - (118)	●	●	●	●	●	●	●	●
A	Deferred Generation Cost Recovery - (117)	●	●	●	●	●	●	●	●
Q	Delivery Capital Recovery - (124)	●	●	●	●				
	Delivery Service Improvement - (108)	●	●	●	●				
Q	Delta Revenue Recovery - (96)	●	●	●	●	●	●	●	●
T	Demand Side Management - (97)	●							
T	Demand Side Management and Energy Efficiency - (115)	●	●	●	●	●	●	●	●
A	Distribution Modernization – (132)	●	●	●	●	●	●	●	●
Q	Distribution Uncollectible - (99)	●	●	●	●	●	●	●	●
Q	Economic Development - (116)	●	●	●	●	●	●	●	●
	Economic Load Response Program - (101)			●	●	●			
	Experimental Critical Peak Pricing - (113)		●	●	●	●			
	Experimental Real Time Pricing - (111)		●	●	●	●			
	Fuel - (105)	●	●	●	●	●	●	●	●
Q	Generation Cost Reconciliation - (103)	●	●	●	●	●	●	●	●
	Generation Service - (114)	●	●	●	●	●	●	●	●
T	Government Directives Recovery – (126)	●	●	●	●	●	●	●	●
	Grandfathered Contract - (94)		●	●	●	●			
	Hospital Net Energy Metering - (87)		●	●	●	●			
Q	Line Extension Cost Recovery - (107)	●	●	●	●	●	●	●	●
	Net Energy Metering - (93)	●	●	●	●	●			
Q	Non-Distribution Uncollectible - (110)	●	●	●	●	●	●	●	●
A	Non-Market-Based Services - (119)	●	●	●	●	●	●	●	●
P	Non-Residential Deferred Distribution Cost Recovery - (121)		●	●	●	●	●	●	●
A	Ohio Renewable Resources - (129)	●	●	●	●	●	●	●	●
A	Peak Time Rebate Program (88)	●							
T	Phase-In Recovery (125)	●	●	●	●	●	●	●	●
Q	PIPP Uncollectible - (109)	●	●	●	●	●	●	●	●
	Reasonable Arrangement - (98)		●	●	●	●			
P	Residential Deferred Distribution Cost Recovery - (120)	●							
	Residential Distribution Credit - (81)	●							
T	Residential Electric Heating Recovery - (122)	●							
	Residential Generation Credit - (123)	●							

SUMMARY RIDER

		RS	GS	GP	GSU	GT	STL	TRF	POL
	School Distribution Credit - (85)		●	●	●				
	State kWh Tax - (92)	●	●	●	●	●	●	●	●
A	Transmission and Ancillary Services - (83)	●	●	●	●	●	●	●	●
P	Universal Service - (90)	●	●	●	●	●	●	●	●

● - Rider is applicable or available to the rate schedules indicated

A - Rider is updated/reconciled annually
Q - Rider is updated/reconciled quarterly
T - Rider is updated/reconciled twice per year
P - Rider is updated/reconciled periodically

Now they require a second page

RIDER NMB
Non-Market-Based Services Rider

NMBC = The amount of the Company's total projected Non-Market-Based Services-related costs for the Computation Period, allocated to each rate schedule.

The Computation Period over which NMB will apply shall be for a 12 month period beginning no later than 75 days after filing, which will be no later than January 15th of each year.

E = Starting June 1, 2012, any net over- or under-collection of the Non-Market-Based Services-related costs, including applicable interest, invoiced during the period from June 1, 2011 to March 31, 2012, allocated to rate schedules. Thereafter, E will be calculated for the 12-month period immediately preceding the Computation Period.

BU = Forecasted billing units for the Computation Period for each rate schedule.

CAT = The Commercial Activity Tax rate as established in Section 5751.03 of the Ohio Revised Code.

NMB charges:

RS (all kWhs, per kWh)	1.3561¢
GS* (per kW of Billing Demand)	$4.4543
GP* (per kW of Billing Demand)	$6.7035
GSU (per kW of Billing Demand)	$5.1925
GT (per kVa of Billing Demand)	$2.8860
STL (all kWhs, per kWh)	0.0000¢
TRF (all kWhs, per kWh)	1.0487¢
POL (all kWhs, per kWh)	0.0000¢

* Separately metered outdoor recreation facilities owned by non-profit, governmental and educational institutions, such as athletic fields, served under Rate GS or GP, primarily for lighting purposes, will be charged per the NMB charge applicable to Rate Schedule POL.

RIDER UPDATES:

The charges contained in this Rider shall be updated and reconciled on an annual basis. The Company will file with the PUCO a request for approval of the Rider NMB charges no later than January 15th of each year, which shall become effective on a service rendered basis no later than 75 days after filing, unless otherwise ordered by the Commission. This Rider is subject to reconciliation, including, but not limited to increases or refunds. Such reconciliation shall be based solely upon the results of audits ordered by the Commission in accordance with the July 18, 2012 Opinion and Order in Case No. 12-1230-EL-SSO, and the March 31, 2016 Opinion and Order in Case No. 14-1297-EL-SSO and upon the Commission's orders in Case No. 18-47-AU-COI.

RIDER DMR
Distribution Modernization Rider

APPLICABILITY:

Applicable to any customer who receives electric service under the Company's rate schedules. This Distribution Modernization Rider (DMR) will be effective for service rendered beginning March 1, 2018. This Rider is not avoidable for customers during the period the customer takes electric generation service from a certified supplier.

The DMR charges will apply, by rate schedule, as follows:

RATE:

RS (all kWhs, per kWh)		0.3314¢
GS (per kW of Billing Demand)	$0.5548	
GS (all kWhs, per kWh)		0.1570¢
GP (per kW of Billing Demand)	$0.7680	
GP (all kWhs, per kWh)		0.1570¢
GSU (per kW of Billing Demand)	$0.6375	
GSU (all kWhs, per kWh)		0.1570¢
GT (per kVA of Billing Demand)	$0.3470	
GT (all kWhs, per kWh)		0.1570¢
STL (all kWhs, per kWh)		0.1570¢
TRF (all kWhs, per kWh)		0.2884¢
POL (all kWhs, per kWh)		0.1570¢

RIDER UPDATES:

The charges contained in this Rider shall be updated on an annual basis. No later than December 1st of each year, the Company will file with the PUCO a request for approval of the Rider charges which, unless otherwise ordered by the PUCO, shall become effective on a service rendered basis on January 1st of each year.

Everyone Tells Me I Pay 10¢, 11¢, 12¢ ....

- The unit cost is
- Total bill $$ divided by energy consumed (kWh)
- So Let's Calculate Your December Bill
- 317,000-kWh;
- 999.3-kW;
- 389-kVAR;
- $30,702.28;
- 9.68¢/kWh $_{AVG}$

- GS Rate Schedule = $7.479 per kW
- GS Riders = $8.234 per kW
- Total GS demand = $15.12/kW
- GS kW = 1,270 x $15.12 = $19,200 (From submeters)
- GP Rate Schedule = $2.405 per kW
- GP Riders = $8.62 per kW
- Total GP demand = $10.98/kW
- GP = 999.3 x $10.98 = $10,970 ≈ 3.46¢/kWh
- Plus $5,700 EUS

The sum is less than the sum the parts

- The demand on one GP meter is less than the total of multiple GS meters
- ***You pay for less units at a lower cost!***
- Savings = $4,500 for December

CEI Billing Structure

- From previous we pay demand (kW)
- GP = 999.3 x \$10.98 = \$10,970 ≈ 3.46¢/kWh
- It is NOT 3.46¢/kWh;
- It IS \$10,970 @ \$10.98 per kW

CEI Billing Structure

- There are Fixed fees that are NOT kWh
- Pay for kVAR @ $0.36/kVAR = $142.32
- Two more flat charges for metering and service fee = $165
- Combo of fixed fees = 0.1¢/kWh
- Equivalent cost per kWh for demand and fixed charges = 3.56¢/kWh
- This is NOT an energy charge
- It IS Demand and Fixed components

So Where Oh Where are the kWh Charges?

- *There are no kWh charges in the GS or GP Rates*
- P4S Generation is 5.18¢/kWh (an interesting factoid: approximately 25% is a demand charge)
- The Riders vary by rate and periodic adjustments
- Approximately .6¢/kWh – 1.5¢/kWh
- December energy Riders - $0.0098
- Total energy charges = $19,520 = $0.0615/kWh (6.15¢/kWh)
- This *IS* the Energy Charge

Demand & Fixed Components		Energy Components	
Rate	$2.405	Generation	$0.0518
Rider - NMB	$6.7035	Rate	$0.0000
Rider - DM	$0.768	Riders	$0.0097
Rider - DCR	$1.1035		
TOTAL	$10.98		$0.0615

Service and Meters $165/month

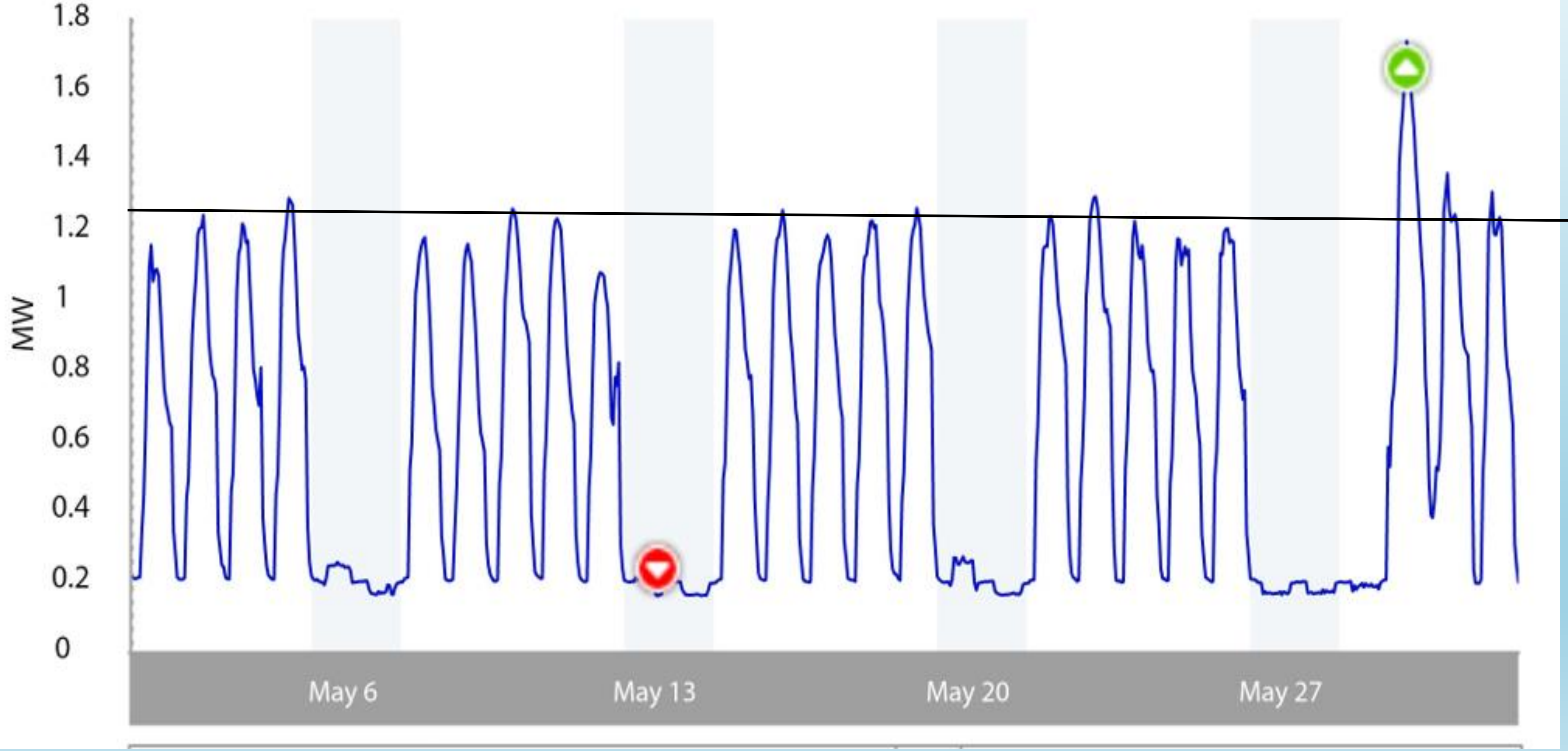
Total not equal to bill

Riders changed slightly on 1/19 and the meter read date is 1/3/19

OK. Now I have a Head Ache. What does it Mean?

- Primary Power Service Saves Money
- Saves more with new construction
- Electric Bill is comprised of Three Components
- Demand – Energy – Fixed
- ***Not all kWhs are created equally***
- Let's look at how the components affect Energy Saving Measures

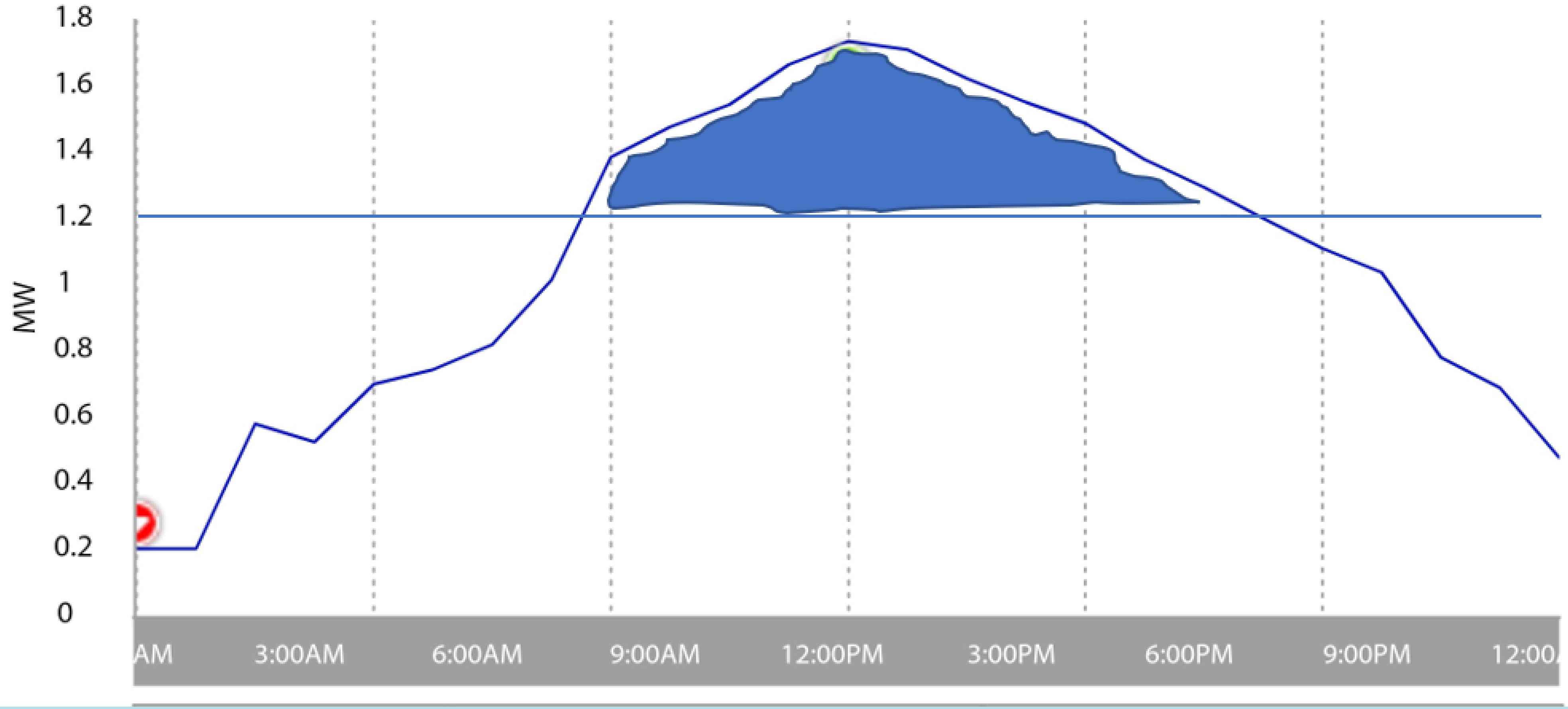
Energy Profiling 05/01/2018 - 05/31/2018



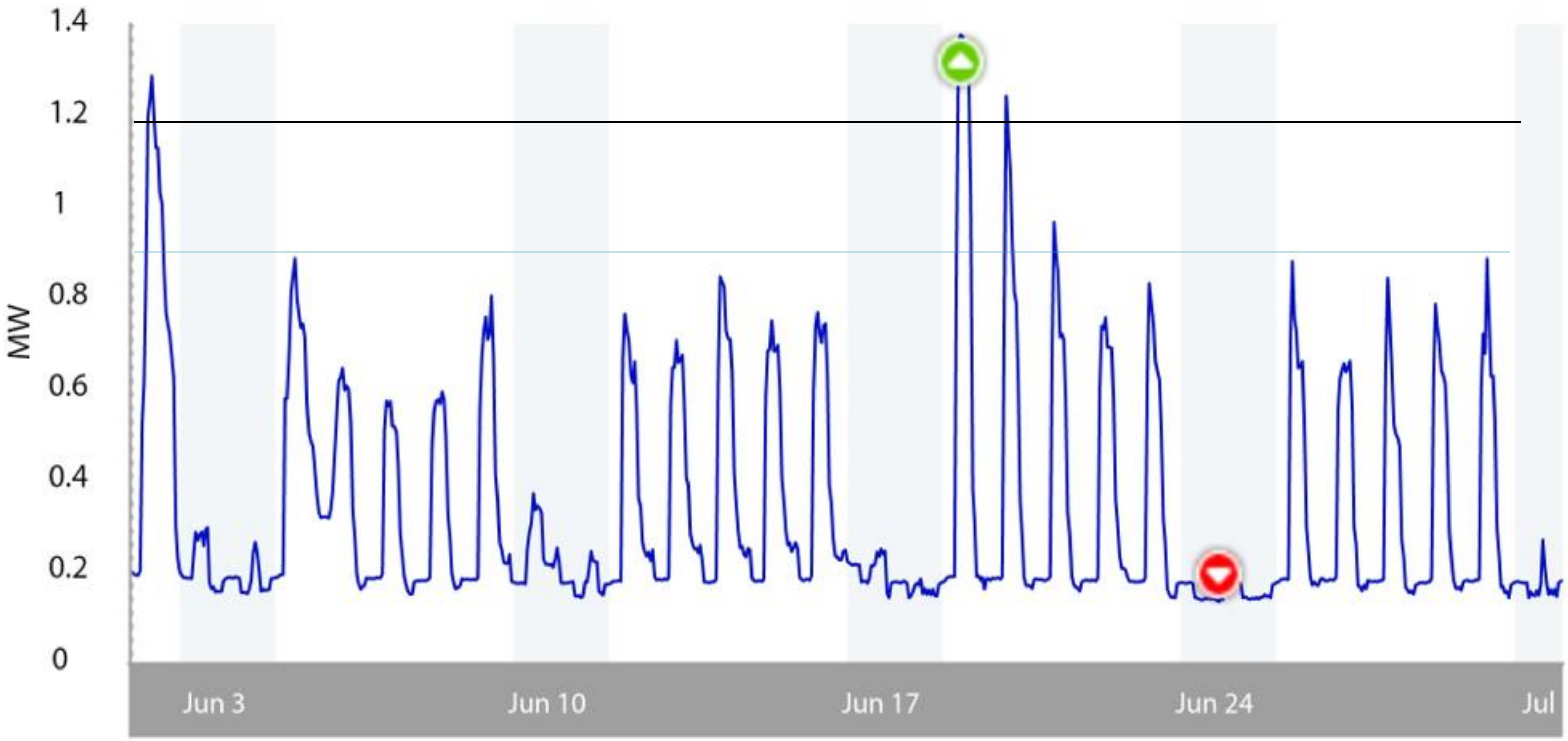
Peak = 1,736kW kWh used from 9:00AM to 7:00PM = 3,200kWh
What is cost of these 10-hours above 1,200kW?
(1736Kw-1200Kw) X $10.98 = $5,885 + 3,200kWh x $0.0615 = $197

$5,885 + $197 = $6,082 → ***$1.90/kWh***
All kWh are not created equal.

Energy·Profiling·05/29/2018·-·05/29/2018¶



Energy·Profiling·06/01/2018·-·06/30/2018¶



How To Reduce Electrical Costs

- Most Schools Have Lowered Consumption – HB Projects
- Most Schools Have Lowered their Generation Rate – P4S
- Few Schools Have Lowered their Distribution Rate
- Generation Rate + Distribution Rate = Total Electrical Rate

The Biggest Savings Opportunity is in Regulated Distribution Rates

Energy Measures

- Energy Efficiency
- Energy Effectiveness
- Energy Conservation
- Demand Side Management
- Renewable Energy
- LEED

Ways to pay

- House Bill 264
- HB 153
- Plan and Specification
- PACE
- Third party own operate and maintain
- Utility incentives and/or Rider avoidance

Energy Efficiency

- Install new equipment
- LED
- VFD
- HVAC

Energy Conservation

- Conservation is a reduction in kWh only
- This means cost savings is energy only of $0.0615/kWh versus the average cost of $0.0968
- Shut things off when not in use
- Unoccupied set back controls

Energy Efficiency

- HB 264 or 153
- Normally hire an ESCO
- Design Build process
- Must borrow money or pay cash
- Most ESCO offer guarantee of savings
- Average cost of power or energy only??
- Pro – better control over selecting subs, team oriented
- Con - Over-emphasis on price can dilute quality

Demand Side Management

- Reduction in demand – kW only
- This means cost savings is demand only of $10.98/kW
  - Limited energy savings
- Control when equipment is on
- Cycle HVAC - air handlers and compressors
- Thermal energy storage
- Run generation

Energy Effectiveness

- Right rate
- Regulated and de-regulated
- Equipment appropriately sized and utilized

Energy Efficiency

- Plan and Specification
- Traditional process of low bid
- Capital expense
- Pro - familiarity
- Con – Get low bidder – change orders – less cooperative

Renewable Energy

- Solar
- Wind
- Heat recovery
- CHP
- Alternative fuels

Solar

- Photovoltaic
- Roof Mount
- Ground Mount
- Trackers
- On Grid
- Off Grid

Solar

- Purchasing Options
- Buy
- Lease
- PPA
- Third Party Ownership

Solar

- Savings
- Does not reduce demand
- Energy Only
- PPA providers tout savings based on average cost
- ***Every kWh is not equal!***

Remember, Every kWh is Not Equal

- kW is your new best energy friend

Contact Information

- Mike Spacek, PE, CEM, LEED AP
  - Member, Operations Manager
  - MikeSpacek@EffectiveUtiltiyServices.com
  - Mike@bestuseofenergy.com
  - 419-602-0503